

2.4 Nitrogen & Sulfur

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.4 Nitrogen & Sulfur
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

In a car exhaust, multiple reactions take place as molecules are mixed with oxygen at high temperatures.

Which of the following reactions has the highest enthalpy change?

- A. $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$
- B. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- C. $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$
- D. $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$

[1 mark]

Question 2

Which reaction using ammonia does **not** involve the nitrogen atom's non-bonding pair of electrons?

- A. $2\text{NH}_3(\text{aq}) + \text{Ag}^+(\text{aq}) \rightarrow [\text{Ag}(\text{NH}_3)_2]^+(\text{aq})$
- B. $2\text{NH}_3(\text{l}) + 2\text{Na}(\text{s}) \rightarrow 2\text{NaNH}_2(\text{s}) + \text{H}_2(\text{g})$
- C. $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$
- D. $\text{NH}_3(\text{g}) + \text{CH}_3\text{I}(\text{g}) \rightarrow \text{CH}_3\text{NH}_3^+\text{I}^-(\text{s})$

[1 mark]

Question 3

A farmer spreads lime on land which has already been treated with an ammonium nitrate fertilizer.

Which reactions will occur in the treated soil?

- 1 $\text{Ca}(\text{OH})_2 + 2\text{NH}_4^+(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{NH}_3 + 2\text{H}_2\text{O}$
- 2 $\text{Ca}(\text{OH})_2 + 2\text{H}^+(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{H}_2\text{O}$
- 3 $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$

- A. 1, 2 and 3
- B. 1 and 2
- C. 2 and 3
- D. 1 only

[1 mark]

Question 4

Use of the data booklet is relevant to this question.

Car exhausts release pollutant gases including oxides of nitrogen.

In cities, when a car travels one kilometre, it releases 0.23 g of an oxide of nitrogen N_xO_y , which occupies 120 cm^3 .

What are the values of x and y ?

(Assume 1 mol of gas molecules occupies 24.0 dm^3 .)

- A. $y = 1, x = 1$
- B. $y = 1, x = 2$
- C. $y = 2, x = 1$
- D. $y = 4, x = 2$

[1 mark]

Question 5

Oxide **X** is a pollutant which contains a non-metallic element **Y**, formed in a car engine.

Further oxidation of **Y** to **Z** occurs in the atmosphere. In this further oxidation, 1 mol of **Y** reacts with 0.5 mol of gaseous oxygen molecules.

Y could be either nitrogen or sulfur.

- 1 The oxidation number of **Y** increases by two from **X** to **Z**.
- 2 **X** is a polar molecule.
- 3 **X** has an unpaired electron in its molecule.

Which statements about **X**, **Y** and **Z** can be correct in all possibilities?

- A. 1, 2 and 3
- B. 1 and 2
- C. 2 and 3
- D. 1 only

[1 mark]

Question 6

Ammonium sulfate in the soil is slowly oxidised by air, producing sulfuric acid, nitric acid and water as the only products.

How many moles of oxygen gas are needed for the complete oxidation of one mole of ammonium sulfate?

- A. 1
- B. 2
- C. 3
- D. 4

[1 mark]

Question 7

Aqueous ammonia is added to aqueous sulphur dioxide until just alkaline.

Which salt is produced in this reaction?

- A. NH_4SO_3
- B. $(\text{NH}_4)_2\text{SO}_3$
- C. NH_4SO_4
- D. $(\text{NH}_4)_2\text{SO}_4$

[1 mark]

Question 8

In a car engine, non-metallic element *P* forms a pollutant oxide *Q*. *Q* can be further oxidised to *R*.

Two students made the following statements.

Student 1: The oxidation number of *P* increases by 1 from *Q* to *R*.

Student 2: The molecule of *Q* contains lone pairs of electrons.

P could be carbon or nitrogen or sulfur.

Which student could be correct if *P* were any of these elements?

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

[1 mark]

Question 9

The gases nitrogen dioxide (NO_2) and sulfur trioxide (SO_3) are required during the formation of acid rain.

In a simulation experiment, the pressure of sulfur trioxide gas was 2 atm in a reaction chamber of 600 cm^3 and kept at a constant temperature of $25\text{ }^\circ\text{C}$. 0.75 moles of nitrogen dioxide is added to the reaction chamber as well as water vapour.

How many moles of nitrogen dioxide will be in the reaction chamber after all the sulfur trioxide has reacted to make acid rain?

- A. 0
- B. 0.25
- C. 0.5
- D. 0.75

[1 mark]

Question 10

After manufacture in a laboratory, ammonia needs to be dried before being collected.

Which drying agents may be used to dry ammonia?

- 1 Calcium oxide, CaO
- 2 Magnesium sulfate, MgSO_4
- 3 Phosphorus (V) oxide, P_4O_{10}

- A. 1 only
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

[1 mark]

Model Answers: Hard

1

The correct answer is **C** because:

- This reaction involves the breaking of one mole of N-N triple bond and one mole of O-O double bond.
- Energy needs to be put into this reaction due to the **strong triple bond in N₂** requiring lots of energy to break.

2

The correct answer is **B** because:

- Sodium **displaces** the hydrogen on the nitrogen atom and the nitrogen retains its lone pair.

A is incorrect as ammonia forms a **complex** with the silver ions, using the nitrogen's lone pair in ammonia.

C is incorrect as the nitrogen **donates its lone pair** to form a **co-ordinate bond** with the **H⁺ ion** from HCl to produce an ammonium ion (NH₄⁺) which forms an ionic compound with Cl⁻.

D is incorrect as the nitrogen **donates its lone pair** to form a **co-ordinate bond** with the **carbon** atom of the **methyl group** to produce an amine (organic compound with an NH₂ group) with an extra hydrogen bound to the nitrogen – this forms an ionic compound with I⁻. *Realistically, another ammonium molecule will come and take the extra hydrogen to form an ammonium ion and an amine.*

3

The correct answer is **A** because:

- The full reaction of lime and ammonium nitrate is:



- This is a standard acid-base reaction of an ammonium compound reacting with a base to produce a salt, ammonia and water.
- **Reaction 1** is the **general reaction** lime and ammonium containing compounds **without the anion** associated with ammonium.
- **Reaction 2** shows calcium hydroxide performing its **role in soil** - to **reduce the acidity** and therefore react with hydrogen ions.
- **Reaction 3** shows a **double displacement** reaction that occurs between aqueous calcium hydroxide and **CO₂ gas** which is present in soil to produce **insoluble calcium carbonate**.

4

The correct answer is **C** because:

- 1 mole of gas occupies 24,000 cm³
- Therefore the moles of nitrogen oxide = $\frac{120}{24,000} = \mathbf{0.005 \text{ moles}}$
- $M_r = \frac{\text{mass}}{\text{moles}}$
- $M_r = \frac{0.23}{0.005} = \mathbf{46 \text{ g/mol}}$
- 1 atom of nitrogen = 14 g/mol and 1 atom of oxygen = 16 g/mol
- $(1 \times 14) + (2 \times 16) = 46$
- Therefore **x = 1** and **y = 2**

5

The correct answer is **B** because:

- If Y is nitrogen: X = NO and Z = NO₂
- If Y is sulfur: X = SO₂ and Z = SO₃

Statement 1 = correct

- The **oxidation number of nitrogen increases by 2**: from +2 in NO to +4 in NO₂
- The oxidation number of **sulfur increases by 2**: from +4 in SO₂ to +6 in SO₃

Statement 2 = correct

- **Oxygen is more electronegative** than both nitrogen and sulfur, meaning the bonding electrons are held closer to oxygen – creating a **dipole**.
- Unlike in CO₂ which is non-polar, SO₂ is **not linear** and therefore the dipole is not evenly distributed resulting in a **polar molecule**.

Statement 3 = incorrect

- NO has an **unpaired electron** on the nitrogen.
- SO₂ has **no** unpaired electrons in the molecule.

6

The correct answer is **D** because:

- The reaction of **ammonium sulfate** and **oxygen** into the products given is:
 - $(\text{NH}_4)_2\text{SO}_4 + \text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + \text{HNO}_3 + \text{H}_2\text{O}$
- The reaction then needs to be balanced:
 - SO₄ is present on both sides and is therefore balanced
 - There are 2 N in (NH₄)₂SO₄ therefore 2 HNO₃ must be produced
 - To correctly balance hydrogens, 2 H₂O must be produced
 - This leaves **4 O₂ required** for reaction with 1 (NH₄)₂SO₄

6

The correct answer is **D** because:

- The reaction of **ammonium sulfate** and **oxygen** into the products given is:
 - $(\text{NH}_4)_2\text{SO}_4 + \text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + \text{HNO}_3 + \text{H}_2\text{O}$
- The reaction then needs to be balanced:
 - SO_4 is present on both sides and is therefore balanced
 - There are 2 N in $(\text{NH}_4)_2\text{SO}_4$ therefore 2 HNO_3 must be produced
 - To correctly balance hydrogens, 2 H_2O must be produced
 - This leaves **4 O_2 required** for reaction with 1 $(\text{NH}_4)_2\text{SO}_4$
- Balanced equation: $(\text{NH}_4)_2\text{SO}_4 + 4\text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + 2\text{HNO}_3 + 2\text{H}_2\text{O}$

7

The correct answer is **B** because:

- This is a **neutralisation reaction**, not a redox reaction, and therefore the **oxidation number** of all species must remain **constant**.
- The oxidation state of nitrogen remains the same in NH_3 and NH_4^+ (-3 in both) as does hydrogen (+1 in both).
- The oxidation number of oxygen in sulfur dioxide is +2 and the oxidation number of sulfur in sulfur dioxide is -4, therefore the product must have the same oxidation numbers this is only the case in the ion SO_3^{2-} .
- Therefore the final product is **$(\text{NH}_4)_2\text{SO}_3$** .

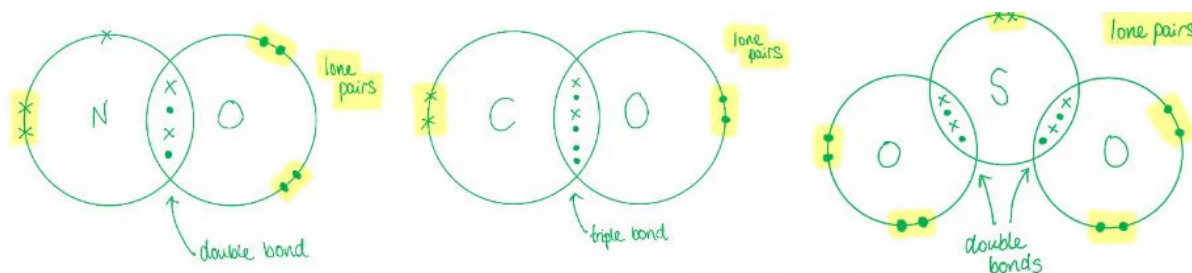
A is incorrect as the **charges** of the ions are **not balanced** in this compound – the charge of an ammonium ion is +1 and the charge of a sulfate ion is 2–.

C is incorrect as the **charges of the ions** are **not balanced** in this compound – the charge of an ammonium ion is +1 and the charge of a sulfate ion is -2. Also, sulfur in SO_4^{2-} has an **oxidation number of -6**, which is different from the -4 state in sulfur dioxide and therefore incorrect.

D is incorrect as sulfur in SO_4^{2-} has an **oxidation number of -6**, which is different from the -4 state in sulfur dioxide and therefore incorrect.

8

The correct answer is **B** because:



Nitrogen

- If P is nitrogen then Q is nitrogen monoxide (NO), which has 3 lone pairs of electrons – 1 on the nitrogen and 2 on the oxygen, therefore **student 2 is correct**.
- Further oxidation of nitrogen monoxide produces nitrogen dioxide.
- The oxidation number of nitrogen in NO is +2 and in NO₂ is +4, therefore **student 1 is wrong** as the oxidation number **increases by 2**.

Carbon

- Carbon would form carbon monoxide which has lone pairs, one on the carbon and one on the oxygen so **student 2 is correct**.
- Carbon monoxide can be oxidised to carbon dioxide (only by a platinum catalyst) and the oxidation number of carbon increases by 2 (from 2 to 4) so **student 1 is wrong**.

Sulfur

- Sulfur will form sulfur monoxide which has 6 lone pairs, 1 on the sulfur and 2 on each oxygen so **student 2 is correct**.
- Sulfur dioxide oxidises to sulfur trioxide in the presence of a nitrogen dioxide catalyst, the oxidation state of sulfur increases by 2 – therefore, **student 1 is wrong**.

9

The correct answer is **D** because:

- Nitrogen dioxide is used as a **catalyst** for the **formation of SO₃** from SO₂ and

oxygen in the natural environment.

- As it is a catalyst it is **not used up** in the reaction and the number of **moles will stay the same**.
- The simulation experiment used premade SO_3 and rather than requiring conversion from sulfur dioxide so the nitrogen dioxide would not be acting as a catalyst anyway!

10

The correct answer is **A** because:

- A drying agent is a chemical that removes water from a mixture by becoming hydrated with water molecules whilst remaining solid.
- Drying agents must not react with the useful product.
- Calcium oxide (also known as quicklime/lime) is used as a drying agent during the manufacture of ammonia.
- Calcium oxide is basic and therefore can be used for basic (and neutral) gases such as ammonia.

B & D is incorrect as when added to water, MgSO_4 dissociates into Mg^{2+} and SO_4^{2-} and **ammonia reacts** with the sulfate ions in this reaction: $2\text{NH}_3 + \text{MgSO}_4 + 2\text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + (\text{NH}_4)_2\text{SO}_4$

C & D is incorrect as **ammonia reacts** with phosphorus oxide in the presence of water in the following reaction: $12\text{NH}_3 + \text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4(\text{NH}_4)_3\text{PO}_4$